

Pragnaya Priyadarshini

Repository: <https://github.com/pragnayap/data260-3170>

Section 0 - Personal configuration

Value	Definition	Mine
SID4	Last four digits of SJSU Student ID	3170
PORT_BASE	$8000 + (\text{SID4} \bmod 900)$	8470
PREFIX	"s" + SID4	s3170
SEED	SID4	3170
VERIFY_SEED	$260000 + \text{SID4}$	263170
DOMAIN_ID	$\text{SID4} \bmod 8$	2 - Municipal Transit Incidents

Run environment

Hardware	MacBook (Intel x86_64), RAM, macOS 15.x
Python	3.12.13
Local model used	sentence-transformers/all-MiniLM-L6-v2 (384-dim, CPU). Part 2 is retrieval-only, so no generative LLM was run - the run log records "LLM is explicitly disabled." HW1/HW2 used qwen3:8b via Ollama.
Tagged commit hash	db0750624e7c36ed8249d4f52a3b2b7964c708ef

Part 1 - FastAPI authentication

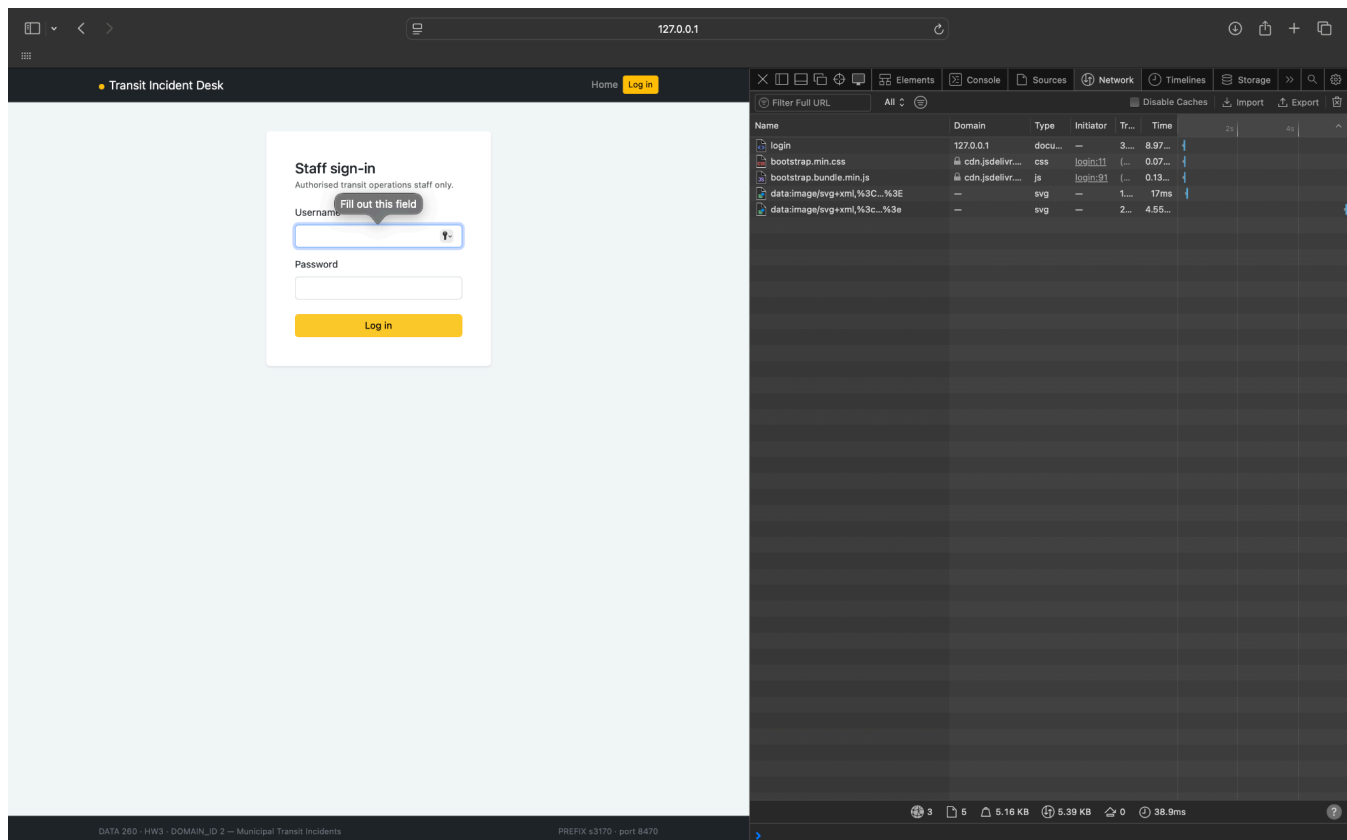
The HW2 application is extended with a session-based login. Every authentication route lives in a single APIRouter in code/web_application/auth.py; app.py keeps the HW2 REST API and adds the middleware wiring. The HW1/HW2 incident form moved from / to /incidents so that / could become the public home page.

Route	Purpose
GET /	Public home page. Links to login when signed out; to dashboard and logout when signed in.
GET /login	Bootstrap login form. Shows an alert on rejected credentials or an expired session.
POST /login	Validates credentials, starts a session, redirects to the dashboard with 303.
GET /dashboard	Protected. Redirects to /login?expired=1 without a live session.
GET /logout	Revokes the session server-side and returns to /.
GET /incidents	The HW1/HW2 incident form, moved off /.
GET /api/incidents...	The HW2 REST API, unchanged.

1.1 Home page (/)

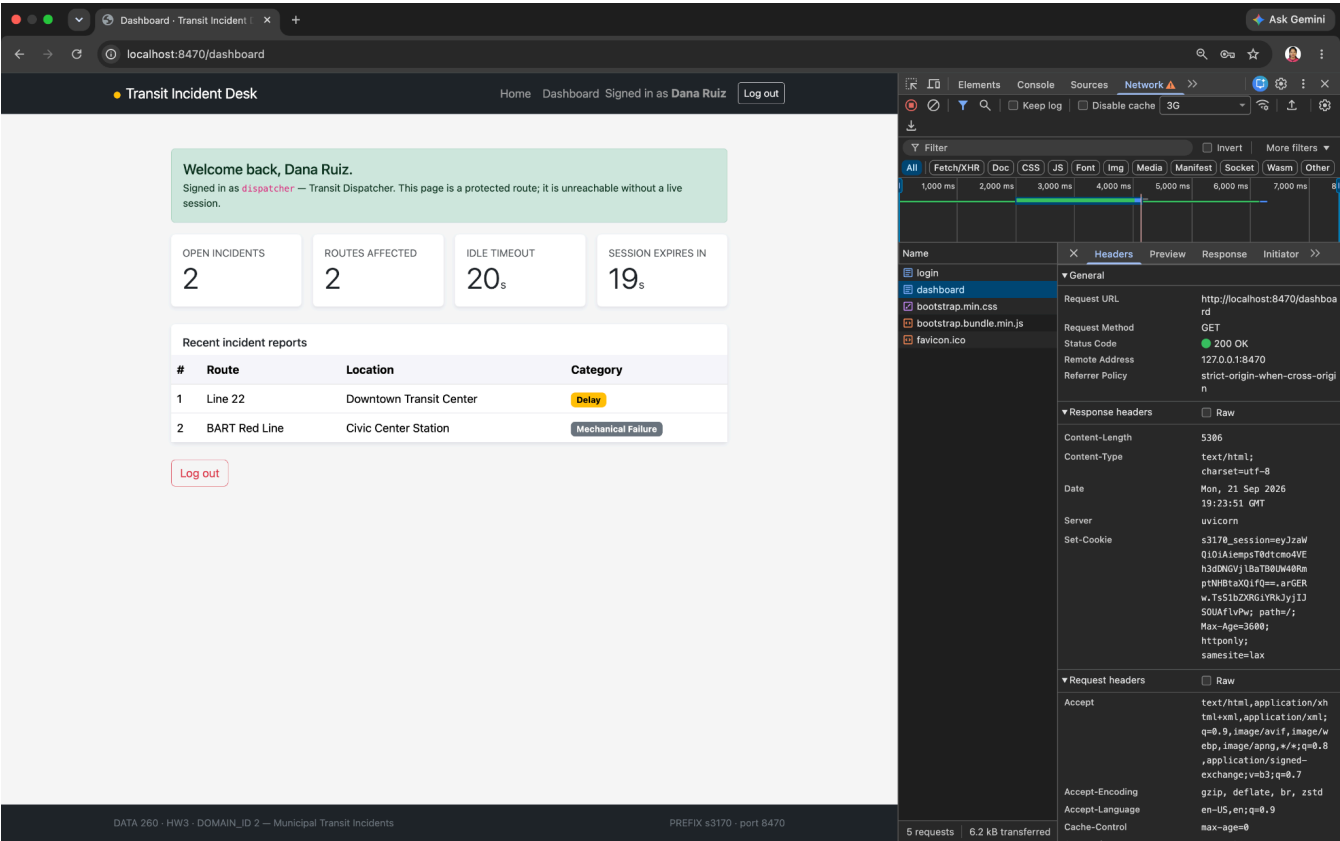
Signed out, the page states that the dashboard is a protected route and offers a single call to action. This screenshot doubles as the evidence for GET /logout, which returns a 303 redirect and lands the browser back here in the signed-out state.

SCREENSHOT: 1.1 - GET / while logged out



Signed in, the same route shows the user's name and role and offers the dashboard and logout.

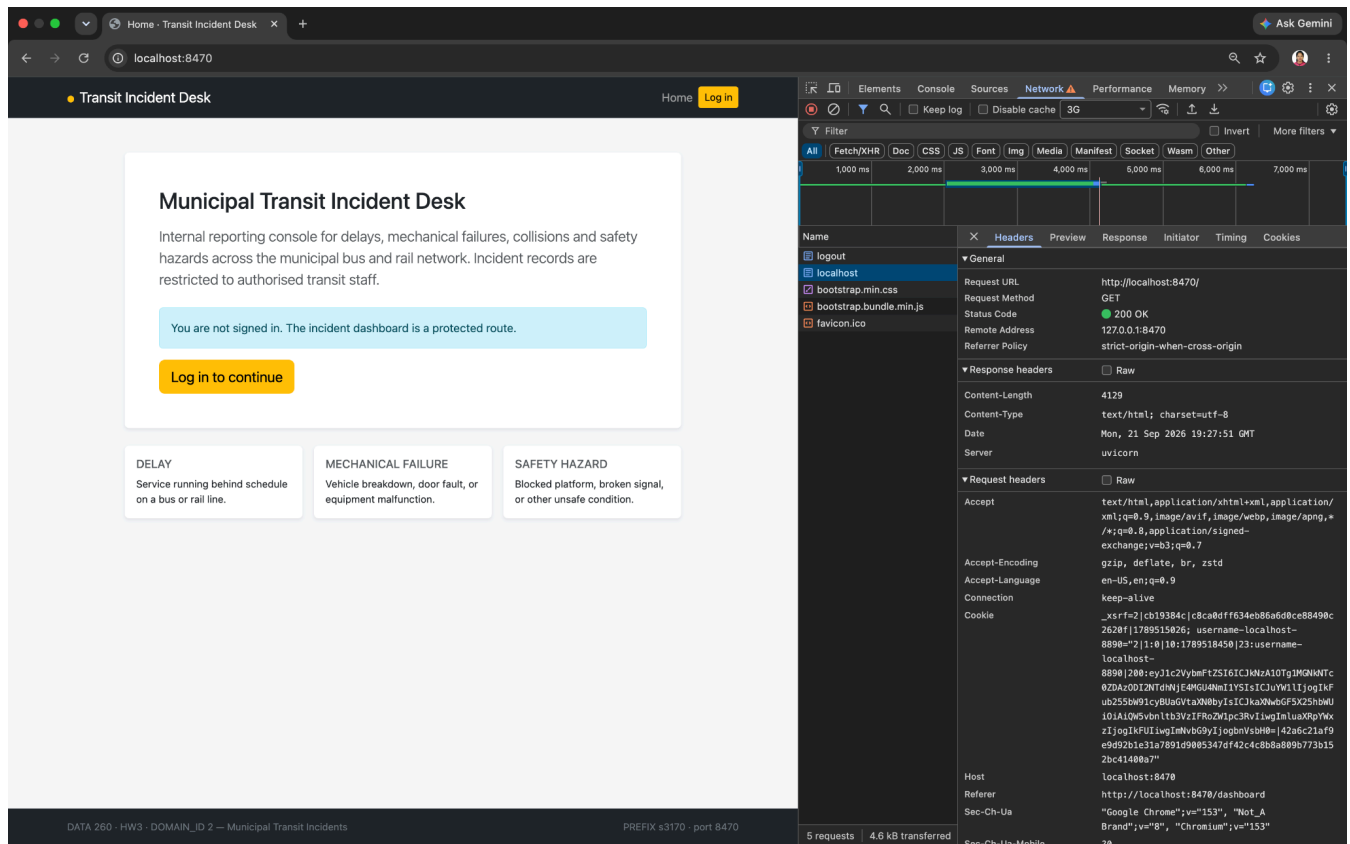
SCREENSHOT: 1.2 - GET / while logged in



1.2 Login page (/login)

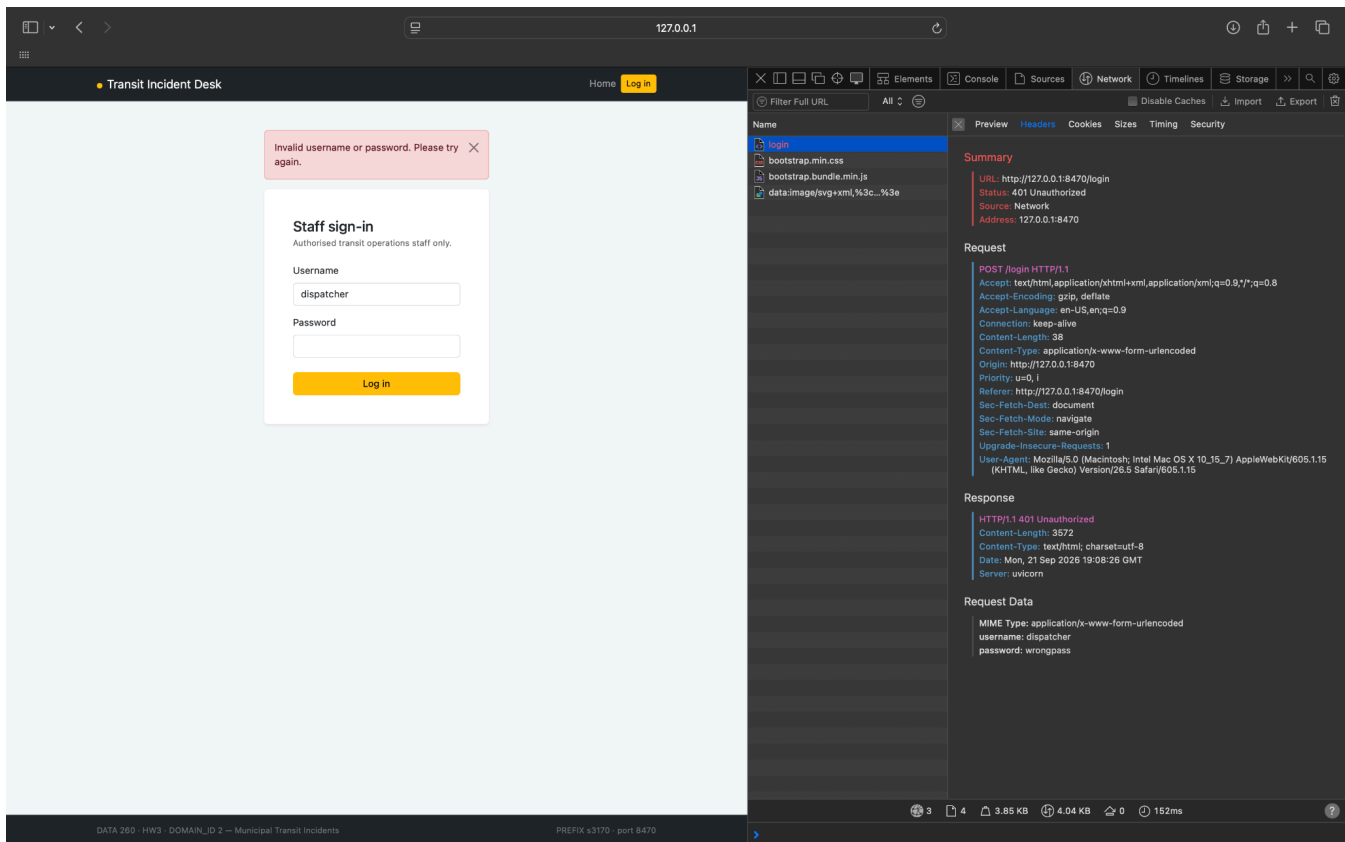
Credentials are checked against a user store holding a per-user salt and a PBKDF2-HMAC-SHA256 derived key at 200,000 iterations. No plaintext password is stored. The comparison uses `hmac.compare_digest`, and an unknown username still runs one hash so a missing account costs the same time as a wrong password - otherwise response timing would reveal which usernames exist.

SCREENSHOT: 1.3 - GET /login



A rejected login returns HTTP 401 with a Bootstrap danger alert, and gives the same message for a wrong username and a wrong password so neither is confirmed.

SCREENSHOT: 1.4 - POST /login with invalid credentials



1.3 Dashboard (/dashboard) - the protected route

The dashboard greets the user by name and shows the idle timeout and the seconds remaining in the current session.

SCREENSHOT: 1.5 - GET /dashboard while logged in

The screenshot shows a web browser at `localhost:8470/dashboard`. The page is titled "Transit Incident Desk" and shows a user signed in as "Dana Ruiz". The dashboard includes a welcome message, statistics for open incidents (2), routes affected (2), idle timeout (20s), and session expiry (19s). It also lists recent incident reports for Line 22 and BART Red Line.

On the right, the browser's developer tools show the network tab with a request to `http://localhost:8470/dashboard`. The response status is 200 OK. The response headers include `Content-Type: text/html; charset=utf-8` and a `Set-Cookie` header with a complex signed cookie value.

1.4 Session management

Starlette's SessionMiddleware stores the session in a signed cookie. The signature makes the cookie tamper-proof but not revocable: a captured copy stays validly signed forever, and `request.session.clear()` on logout only asks the browser to stop sending it.

The first implementation did exactly that, and a test caught it - after logging out, replaying the saved cookie in a fresh client still returned 200 OK on `/dashboard`. Since HW3 requires proving that a logged-out session cannot be reused, the implementation was failing the requirement it was written for.

The fix moves authority from the cookie to the server. Each login mints a random session id with `secrets.token_urlsafe(24)` and registers it in an in-memory `ACTIVE_SESSIONS` dictionary; the cookie carries only that id. Every protected request must pass three gates: a valid signature, an id still in the registry, and an idle window that has not elapsed. Logout and expiry both delete the id, so a replayed cookie fails at the registry lookup even though its signature is still valid.

1.4.1 Cookie attributes

SessionMiddleware always marks the cookie `HttpOnly`; `https_only` adds `Secure` and `same_site` adds `SameSite`. This `Set-Cookie` header was captured automatically by `code/verify_hw03.py` and stored at `reports/hw03/raw/set_cookie_header.txt`.

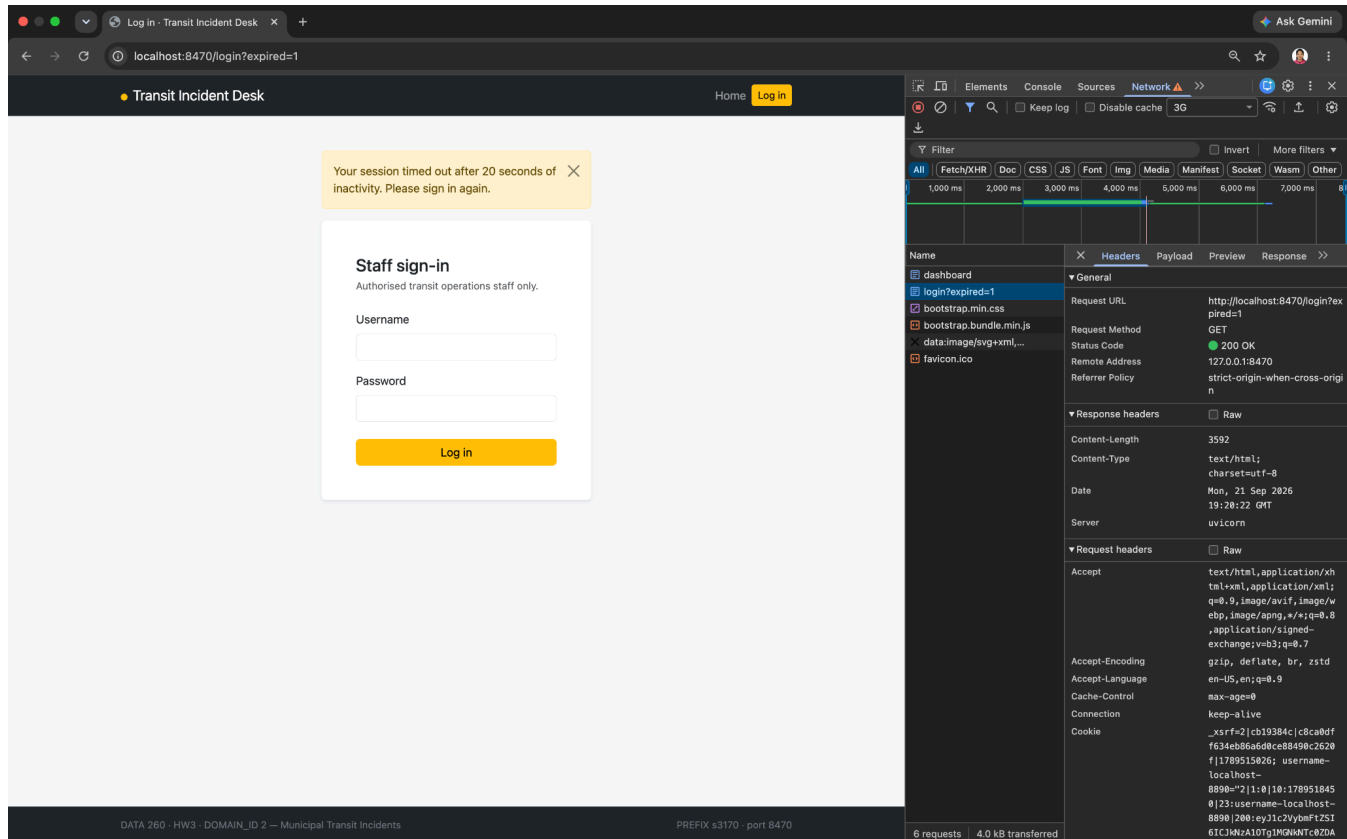
```
3 Set-Cookie: s3170_session=eyJzaWQ10iAiMDE0Q3B1QWVsejJnXzZLQ2NNRUxXTHZ5X2l1Z284NzUifQ==.arF3Tg.ghRaLMytBDxIyXV-2-1xohmVB1M; pa
4
5
```

All three required attributes are present. The cookie payload is only `{"sid": "..."} - the username is not in the cookie at all. Secure stays on by default because browsers treat http://localhost as a secure context, so the cookie still works in local testing.`

1.4.2 Idle timeout

The window is idle time, not a fixed lifetime from login: last_seen is rewritten on every authorised request, so activity keeps a session alive and inactivity kills it. The default is 120 seconds, overridable with S3170_IDLE_TIMEOUT. Past the window the session id is deleted from the registry - revoked, not merely ignored - and the user is redirected to /login?expired=1.

SCREENSHOT: 1.6 - /login?expired=1 after the idle window elapsed



1.4.3 Evidence that expired and logged-out sessions cannot be reused

code/verify_hw03.py starts the app on port 8470 and drives the real flow over HTTP. Three checks cover this requirement, and all three pass:

Check	Result
route:dashboard_anonymous_redirects	303 → /login?expired=1
session:logged_out_cookie_cannot_reach_dashboard	303 - a valid signed cookie replayed after logout is refused
session:idle_timeout_blocks_dashboard	303 → /login?expired=1 after the idle window

SCREENSHOT: 1.7 - terminal output of: make verify-hw03

```
-- Open project in Codex | ritchit -- -zsh      ~/Desktop/DATA 260--Agentic AI/data260-3170 -- -zsh      ~/Desktop/DATA 260--Agentic AI/data260-3170/code/web_application -- -zsh      +

[PASS] bootstrap:alert_on_login
[PASS] bootstrap:cards

Part 1 - live auth flow on port 8470
[PASS] server:port_free -- 8470
[PASS] server:started -- uvicorn appiapp on 8470
[PASS] route:home_200 -- GET / -> 200
[PASS] route:dashboard_anonymous_redirects -- 303 -> /login?expired=1
[PASS] route:login_200 -- GET /login -> 200
[PASS] auth:bad_password_rejected -- -> 401
[PASS] auth:bad_password_shows_alert -- Bootstrap alert rendered on the login page
[PASS] auth:bad_password_issues_no_session
[PASS] auth:unknown_user_same_response -- does not reveal which half was wrong
[PASS] auth:good_login_redirects -- 303 -> /dashboard
[PASS] auth:session_cookie_issued
[PASS] cookie:httponly
[PASS] cookie:secure
[PASS] cookie:samesite
[PASS] cookie:all_three_attributes -- s3170_session=eyJ2eWQ1OIA1MTI1RYU2Z4SHBnQktlX30tdEZNTkE3Qk8yaDhXakhWTWMI1Q==.arGisg.2oARvSo3IsDHohuca15w2_1hkwQ; path=/; M
[PASS] cookie:header_captured_to_file -- reports/hw03/raw/set_cookie_header.txt
[PASS] route:dashboard_authenticated_200 -- GET /dashboard with session -> 200
[PASS] dashboard:greet_user
[PASS] route:logout_redirects -- 303 -> /
[PASS] session:logged_out_cookie_cannot_reach_dashboard -- replayed a valid signed cookie after logout -> 303
[PASS] session:relogin_ok
[PASS] session:live_before_timeout -- -> 200
[PASS] session:idle_timeout_blocks_dashboard -- after 3s idle -> 303 /login?expired=1
[PASS] hw2:api_still_works -- HW2 REST API unaffected by the HW3 changes
[PASS] hw2:form_moved_to_incidents

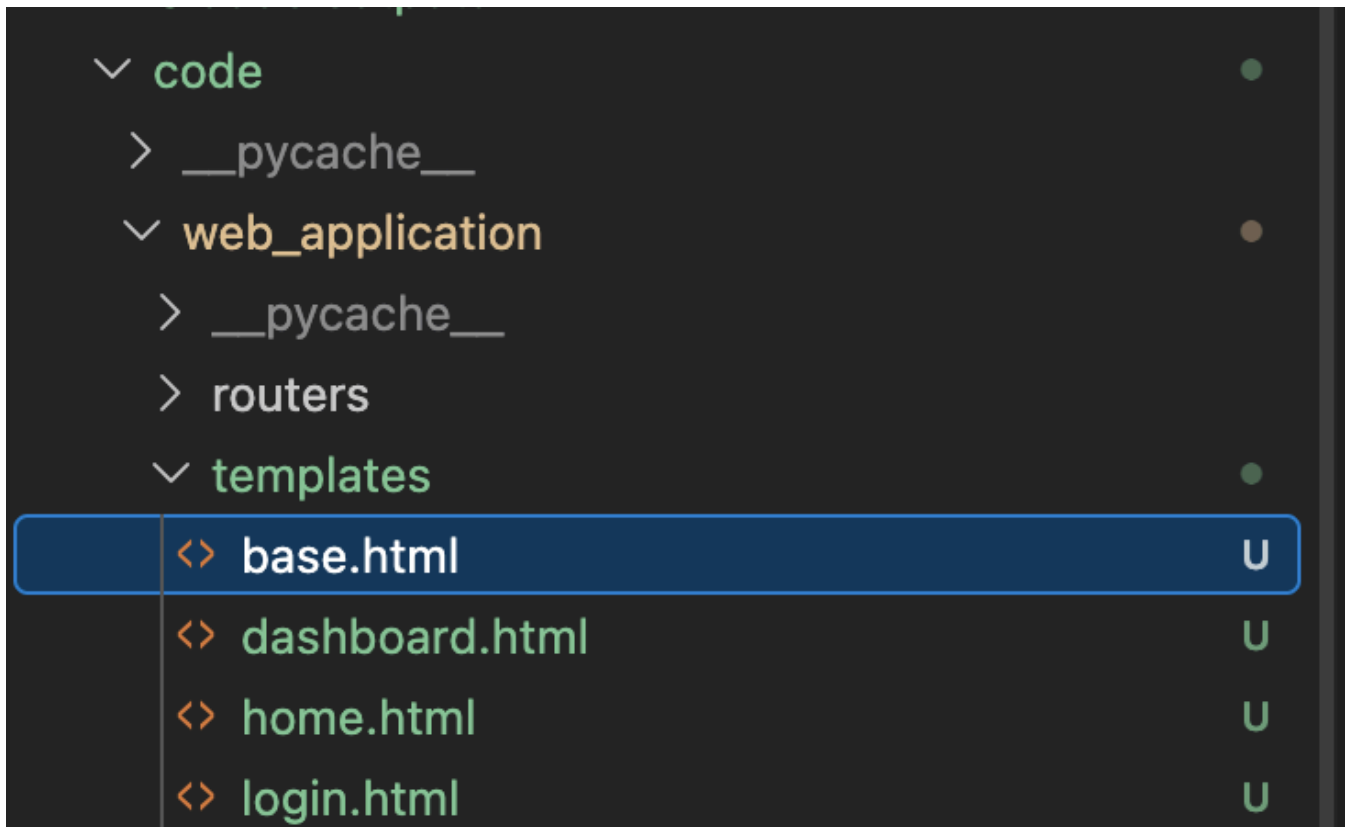
Part 2 - corpus and retrieval artefacts
[PASS] file_exists:fetch_corpus -- /Users/ritchit/Desktop/DATA 260--Agentic AI/data260-3170/code/fetch_corpus.py
[PASS] compiles:fetch_corpus
[PASS] file_exists:run_rag_chunking -- /Users/ritchit/Desktop/DATA 260--Agentic AI/data260-3170/code/run_rag_chunking.py
[PASS] compiles:run_rag_chunking
[PASS] file_exists:summarize_rag_metrics -- /Users/ritchit/Desktop/DATA 260--Agentic AI/data260-3170/code/summarize_rag_metrics.py
[PASS] compiles:summarize_rag_metrics
[PASS] file_exists:questions_yaml -- /Users/ritchit/Desktop/DATA 260--Agentic AI/data260-3170/reports/hw03/questions.yaml
[PASS] file_exists:source_md -- /Users/ritchit/Desktop/DATA 260--Agentic AI/data260-3170/reports/hw03/SOURCES.md
[PASS] file_exists:corpus_manifest -- /Users/ritchit/Desktop/DATA 260--Agentic AI/data260-3170/reports/hw03/CORPUS_MANIFEST.json
[PASS] corpus:present -- 8 files
[PASS] corpus:size_over_200kb -- 4,306,156 bytes (4293.1 KB), floor 284,800
[PASS] corpus:hashes_match_manifest -- 8 documents verified
[PASS] manifest:records_sha256
[PASS] questions:at_least_five -- 5 questions
[PASS] questions:have_expected_answers
[PASS] questions:have_expected_source_files
[PASS] questions:at_least_two_single_source -- 4 marked single_source
[PASS] questions:expected_files_exist_in_corpus -- all present
[PASS] git:questions_committed -- 2026-09-21T19:42:41+00:00
[PASS] git:questions_committed_before_results -- questions 1790019761 vs results 1790019761
[PASS] raw:retrieval_rows -- 75 rows
[PASS] raw:three_techniques -- semantic, sentence, token
[PASS] raw:csv_present
[PASS] raw:graded_embed_model -- embed_model=sentence-transformers/all-MiniLM-L6-v2
[PASS] metrics:writen -- 174 lines
[PASS] metrics:no_placeholders
[PASS] metrics:confident_miss_found -- the assignment requires at least one confidently-wrong retrieval
[PASS] git:hw3_tag_present -- hw3

=====
75 passed, 0 failed, 0 skipped (of 75 checks)
wrote reports/hw03/verification.json
=====
e)
260-3170 on p main [!] via v3.12.13 (.venv) on (us-east-1) took 7s
```

1.5 Styling with Bootstrap

Bootstrap 5.3 is loaded from the CDN with subresource integrity. The layout uses a dark navbar that collapses on narrow screens, cards for the forms and dashboard panels, buttons for actions, and alerts for messages - success on the home page when signed in, danger on a rejected login, warning on an expired session. A responsive table holds the incident list, and the layout is usable down to phone width. Templates are Jinja2: base.html carries the navbar, footer and Bootstrap includes, and home, login and dashboard extend it.

SCREENSHOT: 1.8 - the templates/ directory



1.6 auth.py

Per the assignment, only auth.py is reproduced; app.py, the templates and the stylesheet are in the repository.

part3_step6.txt X schemas.py X verify_hw02.py X report.md X DEMO_CHEATSHEET.md X auth.py X HW1-I

```

data260-3170 > code > web_application > auth.py > ...
245 async def login_submit(
246     record = verify_credentials(username.strip(), password)
247 ):
248     if record is None:
249         # Same message for bad username and bad password -- do not confirm
250         # which half was wrong. 401 so the failure is visible in the network log.
251         return templates.TemplateResponse(
252             "login.html",
253             {
254                 "request": request,
255                 "user": None,
256                 "alert": "Invalid username or password. Please try again.",
257                 "alert_kind": "danger",
258                 "username": username,
259             },
260             status_code=401,
261         )
262
263     # New session id on login: never carry a pre-login session forward.
264     start_session(request, username.strip())
265
266     # 303 forces the browser to GET /dashboard, so a refresh does not re-POST.
267     return RedirectResponse("/dashboard", status_code=303)
268
269 @router.get("/dashboard", response_class=HTMLResponse, include_in_schema=False)
270 async def dashboard(request: Request):
271     """Protected route. Anything without a live session is sent to /login."""
272     user = current_user(request)
273     if user is None:
274         return RedirectResponse("/login?expired=1", status_code=303)
275
276     return templates.TemplateResponse(
277         "dashboard.html",
278         {
279             "request": request,
280             "user": user,
281             "idle_timeout": IDLE_TIMEOUT_SECONDS,
282             "seconds_left": seconds_until_idle_timeout(request),
283             "active_sessions": len(ACTIVE_SESSIONS),
284         },
285     )
286
287 @router.get("/logout", include_in_schema=False)
288 async def logout(request: Request):
289     """Destroy the session and return to the home page.
290
291     end_session() unregisters the sid server-side, so the cookie the browser
292     was holding a moment ago is dead even if it is captured and replayed.
293     """
294     end_session(request)
295     return RedirectResponse("/", status_code=303)

```

st_part3_step6.txt X schemas.py X verify_hw02.py X report.md X DEMO_CHEATSHEET.md X auth.py X HW

data260-3170 > code > web_application > auth.py > ...

```

174 ~/Desktop/DATA 260--Agentic AI/data260-3170 • Contains emphasized items
198     if record is None: # user removed while a session was live
199         end_session(request)
200         return None
201
202     state["last_seen"] = time.time()
203     return {"username": str(state["username"]), **record}
204
205
206 def seconds_until_idle_timeout(request: Request) -> int:
207     """Whole seconds left before this session goes idle-state (for the UI)."""
208     sid = request.session.get("sid")
209     state = ACTIVE_SESSIONS.get(sid) if sid else None
210     if state is None:
211         return 0
212     return max(0, int(IDLE_TIMEOUT_SECONDS - (time.time() - float(state["last_seen"]))))
213
214
215 # --- Routes -----
216
217 @router.get("/", response_class=HTMLResponse, include_in_schema=False)
218 async def home(request: Request):
219     """Public landing page. Shows login, or dashboard+logout when signed in."""
220     return templates.TemplateResponse(
221         "home.html",
222         {"request": request, "user": current_user(request)},
223     )
224
225
226 @router.get("/login", response_class=HTMLResponse, include_in_schema=False)
227 async def login_form(request: Request, expired: int = 0):
228     """The login form. ?expired=1 explains why the user landed back here."""
229     if current_user(request):
230         return RedirectResponse("/dashboard", status_code=303)
231
232     alert = None
233     if expired:
234         alert = (
235             f"Your session timed out after {IDLE_TIMEOUT_SECONDS} seconds of "
236             "inactivity. Please sign in again."
237         )
238     return templates.TemplateResponse(
239         "login.html",
240         {"request": request, "user": None, "alert": alert, "alert_kind": "warning"},
241     )
242
243
244 @router.post("/login", response_class=HTMLResponse, include_in_schema=False)
245 async def login_submit(
246     request: Request,
247     username: str = Form(""),

```

data260-3170 > code > web_application > auth.py > ...

```

140 def start_session(request: Request, username: str) -> str:
144     end_session(request)
145
146     sid = secrets.token_urlsafe(24)
147     now = time.time()
148     ACTIVE_SESSIONS[sid] = {"username": username, "last_seen": now, "login_time": now}
149
150     request.session.clear()
151     request.session["sid"] = sid
152     return sid
153
154
155 def end_session(request: Request) -> None:
156     """Revoke this request's session server-side and clear its cookie."""
157     sid = request.session.get("sid")
158     if sid:
159         ACTIVE_SESSIONS.pop(sid, None)
160     request.session.clear()
161
162
163 def purge_idle_sessions() -> int:
164     """Drop every registered session past the idle window. Returns how many."""
165     cutoff = time.time() - IDLE_TIMEOUT_SECONDS
166     stale = [sid for sid, s in ACTIVE_SESSIONS.items() if s["last_seen"] < cutoff]
167     for sid in stale:
168         del ACTIVE_SESSIONS[sid]
169     return len(stale)
170
171
172 # --- Session helpers -----
173
174 def current_user(request: Request) -> dict[str, str] | None:
175     """The logged-in user for this request, or None.
176
177     Three gates, in order: a signed cookie carrying a sid, that sid still being
178     registered server-side, and the idle window not yet elapsed. Failing any of
179     them revokes the session, which is what makes a logged-out or expired
180     cookie unusable rather than merely stale -- replaying the raw cookie value
181     fails at the registry lookup even though its signature is still valid.
182     """
183     sid = request.session.get("sid")
184     if not sid:
185         return None
186
187     state = ACTIVE_SESSIONS.get(sid)
188     if state is None: # logged out, expired, or never issued here
189         request.session.clear()
190         return None
191
192     if time.time() - float(state["last_seen"]) > IDLE_TIMEOUT_SECONDS:
193         del ACTIVE_SESSIONS[sid] # revoke, do not merely ignore
194         request.session.clear()

```

st_part3_step6.txt × schemas.py × verify_hw02.py × report.md × DEMO_CHEATSHEET.md × auth.py × HV

data260-3170 > code > web_application > auth.py > ...

```

90  _SEED_PASSWORDS = {
91      "dispatcher": "Transit0ps#2026",
92      "inspector": "SafetyFirst#2026",
93  }
94
95
96  def hash_password(password: str, salt_hex: str) -> str:
97      """Derive the stored password hash. Same function used to seed and to verify."""
98      derived = hashlib.pbkdf2_hmac(
99          "sha256",
100         password.encode("utf-8"),
101         bytes.fromhex(salt_hex),
102         PBKDF2_ITERATIONS,
103     )
104     return derived.hex()
105
106
107  def _seed_hashes() -> None:
108      """Compute the stored hashes once at import time from the seed passwords."""
109      for username, plaintext in _SEED_PASSWORDS.items():
110          record = USERS[username]
111          record["password_hash"] = hash_password(plaintext, record["salt"])
112
113
114  _seed_hashes()
115
116
117  def verify_credentials(username: str, password: str) -> dict[str, str] | None:
118      """Return the user record on a correct username+password, else None."""
119      record = USERS.get(username)
120      if record is None:
121          # Still run a hash so a missing username costs the same time as a wrong
122          # password -- otherwise response timing leaks which usernames exist.
123          hash_password(password, "00" * 8)
124          return None
125
126      candidate = hash_password(password, record["salt"])
127      if not hmac.compare_digest(candidate, record["password_hash"]):
128          return None
129      return record
130
131
132  # --- Server-side session registry -----
133  #
134  # sid -> {"username": str, "last_seen": float, "login_time": float}
135  # Present here means "live". Absent means logged out, expired, or forged.
136
137  ACTIVE_SESSIONS: dict[str, dict[str, float | str]] = {}
138
139
140  def start_session(request: Request, username: str) -> str:

```

```
st_part3_step6.txt X schemas.py X verify_hw02.py X report.md X DEMO_CHEATSHEET.md X auth.py X HW1-Pr
data260-3170 > code > web_application > auth.py > ...
38 from __future__ import annotations
39
40 import hashlib
41 import hmac
42 import os
43 import secrets
44 import time
45 from pathlib import Path
46
47 from fastapi import APIRouter, Form, Request
48 from fastapi.responses import HTMLResponse, RedirectResponse
49 from fastapi.templating import Jinja2Templates
50
51 # --- Configuration -----
52
53 # Idle window in seconds. Deliberately short so the timeout is demonstrable in
54 # a screenshot without waiting around; override with the env var for normal use.
55 IDLE_TIMEOUT_SECONDS = int(os.environ.get("S3170_IDLE_TIMEOUT", "120"))
56
57 PBKDF2_ITERATIONS = 200_000
58
59 TEMPLATES_DIR = Path(__file__).parent / "templates"
60 templates = Jinja2Templates(directory=str(TEMPLATES_DIR))
61
62 router = APIRouter(tags=["auth"])
63
64
65 # --- Demo user store -----
66 #
67 # A dict stands in for a user table. Storing the salt and the derived key (both
68 # hex) is exactly what a real users table would hold; the plaintext passwords
69 # are only in the docstring below so the grader can actually log in.
70 #
71 #     dispatcher / TransitOps#2026
72 #     inspector  / SafetyFirst#2026
73
74 USERS: dict[str, dict[str, str]] = {
75     "dispatcher": {
76         "display_name": "Dana Ruiz",
77         "role": "Transit Dispatcher",
78         "salt": "9f2c1a7d4b6e8053",
79         # pbkdf2_hmac('sha256', b'TransitOps#2026', bytes.fromhex(salt), 200000)
80         "password_hash": "", # filled in by _seed_hashes() at import time
81     },
82     "inspector": {
83         "display_name": "Marcus Webb",
84         "role": "Safety Inspector",
85         "salt": "3e7b0d5c1f9a4628",
86         "password_hash": "",
87     },
88 }
```

Part 2 - Comparing three LlamaIndex chunking techniques

Three chunking techniques are built over the same corpus, each with its own in-memory vector index, and compared on retrieval quality for a fixed set of domain questions. No LLM is involved: nothing is generated, the only model is the sentence embedding model, and the only operation is nearest-neighbour search over chunk embeddings. The run log records "LLM is explicitly disabled." That is also what makes the run deterministic - re-running reproduces the same scores exactly.

2.1 Environment and setup

This machine is an Intel Mac, and PyTorch dropped x86_64 macOS support after 2.2.2, so everything sitting on torch is held back to match. faiss-cpu installs per the assignment setup list, but nothing here calls it - the handout specifies an in-memory vector store, so the pipeline uses LlamaIndex's SimpleVectorStore throughout.

```
from __future__ import annotations

import hashlib
import hmac
import os
import secrets
import time
from pathlib import Path

from fastapi import APIRouter, Form, Request
from fastapi.responses import HTMLResponse, RedirectResponse
from fastapi.templating import Jinja2Templates

# --- Configuration -----
+ # PyYAML==6.0.2
# a screenshot without waiting around; override with the env var for normal use.
IDLE_TIMEOUT_SECONDS = int(os.environ.get("S3170_IDLE_TIMEOUT", "120"))

PBKDF2_ITERATIONS = 200_000

TEMPLATES_DIR = Path(__file__).parent / "templates"
templates = Jinja2Templates(directory=str(TEMPLATES_DIR))

router = APIRouter(tags=["auth"])
```

```
3  --- technique: token | question: q1_sacramento_probable_cause -----
4  Q: What did the NTSB determine was the probable cause of the collision between two Sacramento
5  | Regional Transit District light rail trains?
6  query embedding dim : 384
7  query first 8 values: [ 0.000452, 0.003639, 0.033929, 0.087000, -0.011260, -0.013632, -0.030287, 0.089325]
8  expected source      : ntsb_rir2207_sacramento_light_rail_collision.pdf
9
10 rank store_score cosine_sim chunk_len hit source_file preview
11 -----
12 1 0.834981 0.834981 816 YES ntsb_rir2207_sacramento_light_rail_collision Railroad Investigation Report NTSB/RIR-22/07 31 Appendixes Appendix A:
13 2 0.756183 0.756183 675 YES ntsb_rir2207_sacramento_light_rail_collision Railroad Investigation Report NTSB/RIR-22/07 28 3.2 Probable Cause The
14 3 0.753312 0.753312 722 YES ntsb_rir2207_sacramento_light_rail_collision Had the improvement plan not ended, SacRT management would have been r
15 4 0.747894 0.747894 670 YES ntsb_rir2207_sacramento_light_rail_collision Railroad Investigation Report NTSB/RIR-22/07 6 later, 50 yards south o
16 5 0.733110 0.733110 343 YES ntsb_rir2207_sacramento_light_rail_collision Railroad Investigation Report NTSB/RIR-22/07 30 This recommendation is
17
18 query vector shape      : (384,)
19 stacked doc vectors shape : (5, 384)
20 retrieval latency       : 80.557 ms
```

2.2 Dataset

Tiny Shakespeare was used only as a warm-up to confirm the three parsers and the in-memory index work (make warmup). The graded dataset is a corpus of local snapshots of public federal documents about transit safety and transit incidents, assembled for DOMAIN_ID 2.

The mix is deliberate. Regulation (eCFR), rulemaking preamble (Federal Register) and incident narrative (NTSB) are three different genres of prose - steady definitional clauses, long argumentative discussion, and dated factual narrative - and they chunk very differently. That contrast is what the comparison measures.

Local file	Source	Bytes	SHA-256
ecfr_49cfr673_agency_safety_plans.txt	49 CFR Part 673 - Public Transportation Agency Safety Pl...	38,152	7abb2e0208...
ecfr_49cfr674_state_safety_oversight.txt	49 CFR Part 674 - State Safety Oversight	38,462	f57a0b3665...
fr_2024_ptasp_final_rule.txt	Public Transportation Agency Safety Plans - Final Rule (...)	370,003	2bf7cc88b5...

Local file	Source	Bytes	SHA-256
fr_2023_ptasp_proposed_rule.txt	Public Transportation Agency Safety Plans - Notice of Pr...	114,254	94d4df7ada...
fr_2024_national_transit_safety_plan.txt	National Public Transportation Safety Plan (2024)	71,612	915b660326...
fr_2024_general_directive_24_1_assaults.txt	General Directive 24-1: Required Actions Regarding Assau...	125,339	ec2cb20a91...
ntsb_rir2207_sacramento_light_rail_collision.pdf	NTSB/RIR-22/07 - Collision Between Sacramento Regional T...	1,616,498	37df3d0696...
ntsb_rir2315_wmata_rosslyn_derailment.pdf	NTSB/RIR-23/15 - Derailment of Washington Metropolitan A...	2,021,836	249c0d2c3f...

Total: 4,396,156 bytes (4293.1 KB) against a 200 KB floor; 948,734 characters of extracted text across 105 document parts. Full URLs, access dates and complete hashes are in reports/hw03/SOURCES.md and CORPUS_MANIFEST.json. Access date for every document: 2026-09-21.

Two documents are PDFs. NTSB publishes RIR2315.pdf with AES encryption and an empty user password, which pypdf will not open without the cryptography package - pinned in requirements.txt; without it the run fails on that file.

2.3 Questions

Five questions were written and committed before any retrieval was run, so the expected answers could not be adjusted after seeing the output. Every expected answer was taken from the source document itself, not from a model's summary of it. Four of the five depend on information in only one document; the assignment requires at least two.

q5 is deliberately the hardest: the 2023 proposed rule states the same 0.75 percent set-aside in near-identical language to the 2024 final rule, so top-scoring chunks are expected from the wrong document.

```

_step6.txt X schemas.py X verify_hw02.py X report.md X DEMO_CHEATSHEET.md X ! questions.yaml M HW1-Pragnaya-Priyadarshini X
data260-3170 > reports > hw03 > ! questions.yaml
19  questions:
56  - id: q3_two_hour_notification
68      expected_source_file: ecfr_49cfr674_state_safety_oversight.txt
69      single_source: true
70      notes: >-
71          49 CFR 674.33. Part 673 is about safety plans and never states this
72          deadline, so a retrieval that lands in Part 673 is confidently wrong.
73
74  - id: q4_assault_directive_deadline
75      question: >-
76          What does FTA General Directive 24-1 require transit agencies to do about
77          assaults on transit workers, and by when must they respond?
78      expected_answer: >-
79          Agencies must conduct a safety risk assessment of assaults on transit
80          workers using their existing SMS processes (an assessment completed in the
81          preceding 12 months may be reused), identify safety risk mitigations -
82          with the joint labor-management Safety Committee involved for large
83          urbanized area providers - and report to FTA how they are assessing,
84          mitigating and monitoring that risk. Responses were due 26 December 2024,
85          90 days after the directive was issued.
86      expected_source_file: fr_2024_general_directive_24_1_assaults.txt
87      single_source: true
88      notes: >-
89          The shortest document in the corpus. Assaults are also discussed at length
90          in the 2024 final rule and the 2023 proposed rule, so the wrong-document
91          pull is very likely here.
92
93  - id: q5_section_5307_safety_set_aside
94      question: >-
95          What share of its section 5307 funds must a large urbanized area transit
96          provider spend on safety-related projects under the 2024 PTASP final rule?
97      expected_answer: >-
98          At least 0.75 percent of the provider's section 5307 funds must be
99          allocated to safety-related projects, directed towards the areas where
100          the agency missed its safety performance targets.
101      expected_source_file: fr_2024_ptasp_final_rule.txt
102      single_source: false
103      notes: >-
104          Deliberately the hardest of the five. The 2023 proposed rule discusses the
105          same set-aside in near-identical language, so top-scoring chunks are
106          expected to come from the NPRM rather than the final rule. This is the
107          question most likely to surface the required confidently-scored miss.
108
109  # Optional extras, added for the "does the best technique change across queries?"
110  # part of the write-up. Not counted in the five graded questions.
111  extra_questions:
112
113  - id: x1_ptasp_effective_date
114      question: What is the effective date of the 2024 Public Transportation Agency Safety Plans final rule?
115      expected_answer: 13 May 2024.
116      expected_source_file: fr_2024_ptasp_final_rule.txt

```

```
_step6.txt x schemas.py x verify_hw02.py x report.md x DEMO_CHEATSHEET.md x ! questions.yaml M HW1-Pragnaya-Priyadarshini x
data260-3170 > reports > hw03 > ! questions.yaml
14
15 corpus: data/corpus
16 top_k: 5
17 embed_model: sentence-transformers/all-MiniLM-L6-v2
18
19 questions:
20
21 - id: q1_sacramento_probable_cause
22   question: >-
23     What did the NTSB determine was the probable cause of the collision
24     between two Sacramento Regional Transit District light rail trains?
25   expected_answer: >-
26     SacRT's weak administrative controls, which allowed a transportation
27     supervisor to authorize a high-speed test train onto the mainline without
28     knowing the location of passenger train 9 on the same track. Contributing
29     to the collision was senior management's failure to assess that
30     supervisor's competency in the combined controller-and-dispatcher role on
31     the evening shift.
32   expected_source_file: ntsb_rir2207_sacramento_light_rail_collision.pdf
33   single_source: true
34   notes: >-
35     The accident occurred 22 August 2019 near milepost 6.72 on the blue line;
36     thirteen people were transported to hospitals with non-life-threatening
37     injuries. No other corpus document mentions this incident.
38
39 - id: q2_wmata_derailment_cause
40   question: >-
41     Why did Washington Metropolitan Area Transit Authority train 407 derail
42     south of Rosslyn Station?
43   expected_answer: >-
44     An out-of-specification wheelset caused a wheel to depart the rail at a
45     turnout. The wheelset was out of specification because its design allowed
46     the wheels to migrate outward until they exceeded the maximum permitted
47     back-to-back measurement.
48   expected_source_file: ntsb_rir2315_wmata_rosslyn_derailment.pdf
49   single_source: true
50   notes: >-
51     Derailment of 12 October 2021, roughly 166 feet from the Rosslyn Station
52     platform in Arlington, Virginia. A mechanical probable cause, in contrast
53     to the procedural cause in q1 - the two NTSB reports are the most likely
54     pair to be confused with each other.
55
56 - id: q3_two_hour_notification
57   question: >-
58     How soon after a safety event must a rail transit agency notify FTA and
59     its State Safety Oversight Agency, and what kinds of events trigger that
60     notification?
61   expected_answer: >-
62     Within two hours. The triggering events include a fatality, two or more
63     injuries, a derailment, a collision causing one or more injuries, a
64     collision between two rail transit vehicles, a collision causing disabling
65     damage to a rail transit vehicle, an evacuation for life-safety reasons.
```

Pragnaya (15 minutes ago) Ln

2.4 The three chunking techniques

Technique	Parser	Configuration and why
Token	TokenTextSplitter	chunk_size 512, chunk_overlap 64. 512 tokens is roughly a long paragraph of regulatory prose, and MiniLM truncates at 256 word-pieces, so a much larger chunk would be partly invisible to the embedding anyway. The 12.5% overlap means a sentence straddling a boundary still appears whole in one of the two neighbouring chunks.
Semantic	SemanticSplitterNodeParser	buffer_size 2, breakpoint_percentile_threshold 95. A buffer of 2 smooths over the one-sentence asides federal prose is full of, which would otherwise trigger a spurious break. The high percentile keeps breaks rare so chunks stay topical.
Sentence window	SentenceWindowNodeParser	window_size 3. One sentence per node, embedded on the sentence alone, with three sentences either side carried in metadata. Enough context to keep a probable-cause statement with the finding it follows, without pulling in a whole page.

All three index into LlamaIndex's in-memory SimpleVectorStore. The embedding model is the public sentence-transformers/all-MiniLM-L6-v2, 384 dimensions, on CPU.

```
report.md x DEMO_CHEATSHEET.md x run_rag_chunking.py x run_rag_chunking.py (Replace Preview) x
178 dPyYAML==6.0.2l):
196 """
197 from llama_index.core.node_parser import (
198     SemanticSplitterNodeParser,
199     SentenceWindowNodeParser,
200     TokenTextSplitter,
201 )
202
203 token_parser = TokenTextSplitter(chunk_size=512, chunk_overlap=64, separator=" ")
204
205 semantic_parser = SemanticSplitterNodeParser(
206     buffer_size=2,
207     breakpoint_percentile_threshold=95,
208     embed_model=embed_model,
209 )
210
211 sentence_parser = SentenceWindowNodeParser.from_defaults(
212     window_size=3,
213     window_metadata_key="window",
214     original_text_metadata_key="original_sentence",
215 )
216
217 return [
218     ("token", "TokenTextSplitter", token_parser,
219      {"chunk_size": 512, "chunk_overlap": 64}),
220     ("semantic", "SemanticSplitterNodeParser", semantic_parser,
221      {"buffer_size": 2, "breakpoint_percentile_threshold": 95}),
222     ("sentence", "SentenceWindowNodeParser", sentence_parser,
223      {"window_size": 3}),
224 ]
225
226
227 def load_corpus(directory: Path):
228     from llama_index.core import SimpleDirectoryReader
229
230     if not directory.exists() or not any(directory.iterdir()):
231         sys.exit(
232             f"Corpus directory {directory} is empty.\n"
233             "Run: python code/fetch_corpus.py\n"
234             "(that script needs unrestricted network access -- run it in a normal shell)"
235         )
236
237     documents = SimpleDirectoryReader(
238         input_dir=str(directory),
239         filename_as_id=True,
240         required_exts=[".txt", ".pdf", ".md"],
241     ).load_data()
242
243     total_chars = sum(len(d.text) for d in documents)
244     print(f"Loaded {len(documents)} document parts from {directory}")
245     print(f" total extracted text: {total_chars:,} characters")
246     by_file: dict[str, int] = {}
247     for document in documents:
```

Ln 183,

2.5 The retrieval-only function

For each query the function computes the query embedding and prints its dimension and first 8 values, retrieves top-k nodes, and for each prints the store similarity score, an explicitly recomputed cosine similarity, the chunk

length and a 160-character preview, followed by the shapes of the query vector and the stacked document vectors.

The cosine column is not copied from the store score - the retrieved chunks are re-embedded and the cosine computed against the query embedding. That re-embedding uses MetadataMode.EMBED rather than the bare node text, because LlamaIndex embeds a node together with whatever metadata is not on its exclusion list. With the bare text the two numbers disagreed by as much as 0.18 for sentence-window chunking; that was never a scoring bug, just a different string being embedded. After the fix the two agree to six decimal places for all three techniques, which also confirms the store score really is cosine similarity.

```
data260-3170 > code > run_rag_chunking.py > retrieve_and_report
300 def retrieve_and_report(index, embed_model, technique, question_id, question,
34 /
348     for rank, scored_node in enumerate(results, start=1):
349         node = scored_node.node
350         text = node.get_content(metadata_mode=MetadataMode.NONE)
351         embedded_text = node.get_content(metadata_mode=MetadataMode.EMBED)
352
353         document_vector = np.asarray(
354             embed_model.get_text_embedding(embedded_text), dtype=np.float32
355         )
356         document_vectors.append(document_vector)
357
358         source_file = node.metadata.get("file_name", "?")
359         is_hit = source_file == expected_source
360         window_text = node.metadata.get("window", text)
361         cosine_sim = cosine(query_vector, document_vector)
362
363         rows.append(
364             RetrievalRow(
365                 question_id=question_id,
366                 question=question,
367                 technique=technique,
368                 rank=rank,
369                 store_score=round(float(scored_node.score or 0.0), 6),
370                 cosine_sim=round(cosine_sim, 6),
371                 chunk_len_chars=len(text),
372                 chunk_len_tokens_est=estimate_tokens(text),
373                 embedded_text_len_chars=len(embedded_text),
374                 window_len_chars=len(window_text),
375                 source_file=source_file,
376                 expected_source_file=expected_source,
377                 is_expected_source=is_hit,
378                 node_id=node.node_id,
379                 preview=preview_of(text),
380                 retrieval_latency_ms=round(latency_ms, 3),
381                 embed_dim=int(query_vector.shape[0]),
382             )
383         )
384
385         print(f"rank:>4} {float(scored_node.score or 0.0):>11.6f} {cosine_sim:>10.6f} "
386               f"{len(text):>9} {'YES' if is_hit else 'no':>3} {source_file[:44]:<44} "
387               f"{preview_of(text)[:70]}")
388
389     stacked = np.vstack(document_vectors) if document_vectors else np.empty((0, 0))
390     print()
391     if len(results) < top_k:
392         print(f"note: retriever returned {len(results)} of {top_k} requested "
393               f"(the store drops exact-duplicate chunk embeddings)")
394     print(f"query vector shape      : {tuple(query_vector.shape)}")
395     print(f"stacked doc vectors shape : {tuple(stacked.shape)}")
396     print(f"retrieval latency      : {latency_ms:.3f} ms")
397
```

2.6 Printed retrieval output

Each block shows the technique, the query embedding dimension and first 8 values, the rank table (rank, store_score, cosine_sim, chunk_len, source, preview) and the vector shapes.

SCREENSHOT: 2.1 - Token chunking output

```
--- technique: token | question: q1_sacramento_probable_cause -----
Q: What did the NTSB determine was the probable cause of the collision between two Sacramento
Regional Transit District Light rail trains?
query embedding dim : 384
query first 8 values: [ 0.000452, 0.003639, 0.033929, 0.087000, -0.011260, -0.013632, -0.030287, 0.089325]
expected source      : ntsb_rir2207_sacramento_light_rail_collision.pdf

rank  store_score  cosine_sim  chunk_len  hit  source_file  preview
-----
1      0.834981     0.834981     816  YES  ntsb_rir2207_sacramento_light_rail_collision  Railroad Investigation Report NTSB/RIR-22/07 31 Appendixes Appendix A:
2      0.756183     0.756183     675  YES  ntsb_rir2207_sacramento_light_rail_collision  Railroad Investigation Report NTSB/RIR-22/07 28 3.2 Probable Cause The
3      0.753312     0.753312     722  YES  ntsb_rir2207_sacramento_light_rail_collision  Had the improvement plan not ended, SacRT management would have been r
4      0.747894     0.747894     670  YES  ntsb_rir2207_sacramento_light_rail_collision  Railroad Investigation Report NTSB/RIR-22/07 6 later, 50 yards south o
5      0.733110     0.733110     343  YES  ntsb_rir2207_sacramento_light_rail_collision  Railroad Investigation Report NTSB/RIR-22/07 30 This recommendation is

query vector shape      : (384,)
stacked doc vectors shape : (5, 384)
retrieval latency       : 80.557 ms
```

SCREENSHOT: 2.2 - Semantic chunking output

```
data260-3170 > reports > hw03 > RUN_LOG.txt
=====
137  TECHNIQUE: semantic (SemanticSplitterNodeParser)
138  =====
139
140  chunks produced      : 442
141  avg chunk length     : 2146.5 chars (~390.7 tokens est.)
142  min / max chunk len : 3 / 51993 chars
143  chunking time       : 699.84s
144  embedding time      : 22.87s
145  chunks per file:
146      ecf_r49cfr673_agency_safety_plans.txt      12
147      ecf_r49cfr674_state_safety_oversight.txt    13
148      fr_2023_ptasp_proposed_rule.txt            38
149      fr_2024_general_directive_24_1_assaults.txt 36
150      fr_2024_national_transit_safety_plan.txt    22
151      fr_2024_ptasp_final_rule.txt               126
152      ntsb_rir2207_sacramento_light_rail_collision.pdf 82
153      ntsb_rir2315_wmata_rosslyn_derailment.pdf 113
154
155  --- technique: semantic | question: q1_sacramento_probable_cause -----
156  Q: What did the NTSB determine was the probable cause of the collision between two Sacramento
157  Regional Transit District Light rail trains?
158  query embedding dim : 384
159  query first 8 values: [ 0.000452, 0.003639, 0.033929, 0.087000, -0.011260, -0.013632, -0.030287, 0.089325]
160  expected source      : ntsb_rir2207_sacramento_light_rail_collision.pdf
161
162  rank  store_score  cosine_sim  chunk_len  hit  source_file  preview
163  -----
164      1      0.851699     0.851699     507  YES  ntsb_rir2207_sacramento_light_rail_collision  Railroad Investigation Report NTSB/RIR-22/07 31 Appendixes Appendix A:
165      2      0.770044     0.770044     137  YES  ntsb_rir2207_sacramento_light_rail_collision  —. 2016. Collision of Two Union Pacific Railroad Freight Trains Hoxi
166      3      0.756183     0.756183     679  YES  ntsb_rir2207_sacramento_light_rail_collision  Railroad Investigation Report NTSB/RIR-22/07 28 3.2 Probable Cause The
167      4      0.752387     0.752387     474  YES  ntsb_rir2207_sacramento_light_rail_collision  Railroad Investigation Report NTSB/RIR-22/07 v Executive Summary What
168      5      0.751401     0.751401     404  YES  ntsb_rir2207_sacramento_light_rail_collision  Railroad Investigation Report NTSB/RIR-22/07 20 2. Analysis 2.1 Intro
169
170  query vector shape      : (384,)
171  stacked doc vectors shape : (5, 384)
172  retrieval latency       : 86.130 ms
173
174  --- technique: semantic | question: q2_wmata_derailment_cause -----
175  Q: Why did Washington Metropolitan Area Transit Authority train 407 derail south of Rosslyn Station?
176  query embedding dim : 384
177  query first 8 values: [-0.059289, 0.062475, 0.041049, 0.081998, -0.034537, -0.002001, 0.014964, -0.000480]
178  expected source      : ntsb_rir2315_wmata_rosslyn_derailment.pdf
179
180  rank  store_score  cosine_sim  chunk_len  hit  source_file  preview
181  -----
182      1      0.709910     0.709910     1371  YES  ntsb_rir2315_wmata_rosslyn_derailment.pdf  Railroad Investigation Report RIR-23-15 1 1 Factual Information 1.1 De
183      2      0.706700     0.706700     962  YES  ntsb_rir2315_wmata_rosslyn_derailment.pdf  Derailment of Washington Metropolitan Area Transit Authority Train Nea
184      3      0.696375     0.696375     547  YES  ntsb_rir2315_wmata_rosslyn_derailment.pdf  Railroad Investigation Report RIR-23-15 42 3 Conclusions 3.1 Findings
185      4      0.683272     0.683272     515  YES  ntsb_rir2315_wmata_rosslyn_derailment.pdf  Railroad Investigation Report RIR-23-15 vi Executive Summary What Happ
186      5      0.674413     0.674413     506  YES  ntsb_rir2315_wmata_rosslyn_derailment.pdf  Railroad Investigation Report RIR-23-15 45 Appendixes Appendix A: Inve
```

SCREENSHOT: 2.3 - Sentence-window chunking output

```
data260-3170 > reports > hw03 > RUN_LOG.txt
248
249 =====
250 TECHNIQUE: sentence (SentenceWindowNodeParser)
251 =====
252 chunks produced : 6008
253 avg chunk length : 157.9 chars (~28.3 tokens est.)
254 min / max chunk len : 2 / 3508 chars
255 chunking time : 2.03s
256 embedding time : 510.16s
257 chunks per file:
258   ecfr_49cfr673_agency_safety_plans.txt          217
259   ecfr_49cfr674_state_safety_oversight.txt        229
260   fr_2023_ptasp_proposed_rule.txt                 734
261   fr_2024_general_directive_24_1_assaults.txt      684
262   fr_2024_national_transit_safety_plan.txt         415
263   fr_2024_ptasp_final_rule.txt                    2485
264   ntsb_rir2207_sacramento_light_rail_collision.pdf 554
265   ntsb_rir2315_wmata_rosslyn_derailment.pdf       690
266
267 --- technique: sentence | question: q1_sacramento_probable_cause -----
268 Q: What did the NTSB determine was the probable cause of the collision between two Sacramento
269   Regional Transit District light rail trains?
270 query embedding dim : 384
271 query first 8 values: [ 0.000452, 0.003639, 0.033929, 0.007000, -0.011260, -0.013632, -0.030287, 0.089325]
272 expected source : ntsb_rir2207_sacramento_light_rail_collision.pdf
273
274 rank store_score cosine_sim chunk_len hit source_file preview
275 -----
276 1 0.828693 0.828693 378 YES ntsb_rir2207_sacramento_light_rail_collision Railroad Investigation Report NTSB/RIR-22/07 31 Appendixes Appendix A:
277 2 0.823417 0.823417 479 YES ntsb_rir2207_sacramento_light_rail_collision Railroad Investigation Report NTSB/RIR-22/07 28 3.2 Probable Cause The
278 3 0.797674 0.797674 123 YES ntsb_rir2207_sacramento_light_rail_collision In April 2015, the NTSB released its report on the March 23, 2014, col
279 4 0.795702 0.795702 305 YES ntsb_rir2207_sacramento_light_rail_collision The NTSB concludes that none of the following were factors in this col
280 5 0.794701 0.794701 406 YES ntsb_rir2207_sacramento_light_rail_collision The National Transportation Safety Board determines that the probable
281
282 query vector shape : (384,)
283 stacked doc vectors shape : (5, 384)
284 retrieval latency : 235.088 ms
285
286 --- technique: sentence | question: q2_wmata_derailment_cause -----
287 Q: Why did Washington Metropolitan Area Transit Authority train 407 derail south of Rosslyn Station?
288 query embedding dim : 384
289 query first 8 values: [-0.059289, 0.062475, 0.041049, 0.081998, -0.034537, -0.002001, 0.014964, -0.000480]
290 expected source : ntsb_rir2315_wmata_rosslyn_derailment.pdf
291
292 rank store_score cosine_sim chunk_len hit source_file preview
293 -----
294 1 0.760448 0.760448 402 YES ntsb_rir2315_wmata_rosslyn_derailment.pdf Derailment of Washington Metropolitan Area Transit Authority Train Nea
295 2 0.736947 0.736947 428 YES ntsb_rir2315_wmata_rosslyn_derailment.pdf local time, Washington Metropolitan Area Transit Authority (WMATA) tra
296 3 0.736765 0.736765 273 YES ntsb_rir2315_wmata_rosslyn_derailment.pdf local time, Washington Metropolitan Area Transit Authority (WMATA) tra
297 4 0.724751 0.724751 268 YES ntsb_rir2315_wmata_rosslyn_derailment.pdf 3.2 Probable Cause The National Transportation Safety Board determines
```

2.7 Retrieval quality

Recall@k is the fraction of questions whose expected source file appears among the top-k retrieved chunks. It is document-level recall, not passage-level: a chunk from the right file that does not contain the answer still counts as a hit, which is why the confidently-wrong retrievals are reported separately in 2.8. Every figure below is recomputed from reports/hw03/raw/retrieval_rows.jsonl by code/summarize_rag_metrics.py; nothing is transcribed by hand.

Technique	Chunks	Avg chunk length	Top-1 cosine	Mean@5 cosine	Recall@5	Mean retrieval latency (ms)
Token	544	1,980 chars (~361 tok)	0.7532	0.7193	1.00 (5/5)	93.32
Semantic	442	2,146 chars (~391 tok)	0.7476	0.7158	1.00 (5/5)	92.21
Sentence window	6,008	158 chars (~28 tok)	0.7809	0.7615	1.00 (5/5)	235.14

Chunking cost and spread

Technique	Min chunk	Max chunk	Chunking time (s)	Embedding time (s)
Token	197	3,506	1.88	29.66
Semantic	3	51,993	699.84	22.87
Sentence window	2	3,508	2.03	510.16

2.7.1 Per-question top-1 cosine and document hit

Question	Token top-1	Semantic top-1	Sentence window top-1	Token hit	Semantic hit	Sentence window hit
q1_sacramento_probable_cause	0.8350	0.8517	0.8287	yes	yes	yes
q2_wmata_derailment_cause	0.7183	0.7099	0.7604	yes	yes	yes
q3_two_hour_notification	0.7005	0.6814	0.6914	yes	yes	yes
q4_assault_directive_deadline	0.8057	0.7780	0.8442	yes	yes	yes
q5_section_5307_safety_set_aside	0.7063	0.7172	0.7797	yes	yes	yes

2.8 A confidently scored retrieval that does not contain the answer

Question	Technique	Rank	Cosine	Came from	Should have been	Preview
q3_two_hour_notification	Semantic	1	0.6814	fr_2024_ptasp_final_rule.txt	ecfr_49cfr674_state_safety_oversight.txt	673.29(b), and whether this requirement applies to all relevant hazards or only hazards that meet a determined
q4_assault_directive_deadline	Semantic	5	0.7450	fr_2024_ptasp_final_rule.txt	fr_2024_general_directive_24_1_assaults.txt	\13\ National Public Transportation Safety Plan, 88 FR 34917 (May 31, 2023). https://www.federalregister.gov/d
q5_section_5307_safety_set_aside	Sentence window	4	0.7360	fr_2023_ptasp_proposed_rule.txt	fr_2024_ptasp_final_rule.txt	This includes the requirement that if a large urbanized area provider does not meet one of the safety risk red
q5_section_5307_safety_set_aside	Sentence window	5	0.7307	fr_2024_national_transit_safety_plan.txt	fr_2024_ptasp_final_rule.txt	Applicability Comments: Two commenters expressed that the National Safety Plan and safety performance measurem
q5_section_5307_safety_set_aside	Semantic	2	0.7120	fr_2023_ptasp_proposed_rule.txt	fr_2024_ptasp_final_rule.txt	SUPPLEMENTARY INFORMATION: Table of Contents I. Executive Summary A. Purpose of Regulatory Action B. Statutory
q5_section_5307_safety_set_aside	Token	3	0.6936	fr_2023_ptasp_proposed_rule.txt	fr_2024_ptasp_final_rule.txt	of Federal financial assistance under 49 U.S.C. 5307 or a rail transit agency. Transit Asset Management Plan m
q5_section_5307_safety_set_aside	Token	4	0.6879	fr_2023_ptasp_proposed_rule.txt	fr_2024_ptasp_final_rule.txt	be more likely to adopt as a result of developing risk reduction programs or explicitly considering bus collis
q5_section_5307_safety_set_aside	Semantic	4	0.6804	fr_2023_ptasp_proposed_rule.txt	fr_2024_ptasp_final_rule.txt	L. 114-94; December 4, 2015). To implement the requirements of 49 U.S.C. 5329(d), FTA

Question	Technique	Rank	Cosine	Came from	Should have been	Preview
						issued a final rule on J
q3_two_hour_notification	Sentence window	3	0.6800	fr_2024_national_transit_safety_plan.txt	ecfr_49cfr674_state_safety_oversight.txt	One commenter recommended that FTA allow individual transit agencies to define what events will be included in
q3_two_hour_notification	Sentence window	4	0.6767	fr_2024_ptasp_final_rule.txt	ecfr_49cfr674_state_safety_oversight.txt	Similarly, FTA has not established a time frame for informing transit workers of hazards, associated safety ri
q3_two_hour_notification	Token	2	0.6718	fr_2023_ptasp_proposed_rule.txt	ecfr_49cfr674_state_safety_oversight.txt	safety performance targets based on the safety performance measures established under the National Public Tran
q3_two_hour_notification	Sentence window	5	0.6694	fr_2024_national_transit_safety_plan.txt	ecfr_49cfr674_state_safety_oversight.txt	FTA disagrees with the commenter who suggested transit agencies should define what events to include in the ma
q3_two_hour_notification	Semantic	2	0.6650	ntsb_rir2315_wmatarosslyn_derailment.pdf	ecfr_49cfr674_state_safety_oversight.txt	The National Public Transportation Safety Plan also includes guidance on how to set safety performance measure
q3_two_hour_notification	Token	4	0.6603	fr_2024_national_transit_safety_plan.txt	ecfr_49cfr674_state_safety_oversight.txt	until FTA establishes related performance [[Page 25318]] measures through the National Safety Plan (https://www
q3_two_hour_notification	Token	5	0.6566	fr_2024_ptasp_final_rule.txt	ecfr_49cfr674_state_safety_oversight.txt	Comments: One commenter requested that FTA clarify when a transit agency must communicate hazards relevant to
q3_two_hour_notification	Semantic	4	0.6248	ecfr_49cfr673_agency_safety_plans.txt	ecfr_49cfr674_state_safety_oversight.txt	5311 unless it operates a rail fixed guideway public transportation system. § 673.3 Policy. The Federal Transi

Question	Technique	Rank	Cosine	Came from	Should have been	Preview
q3_two_hour_notification	Semantic	5	0.6190	fr_2024_national_transit_safety_plan.txt	ecfr_49cfr674_state_safety_oversight.txt	L. 117-58). On February 5, 2016, FTA first published a Federal Register notice (81 FR 6372) seeking comment on

Why the embedding scored it that highly

The single rank-1 miss is the one that matters, because it is the chunk a naive RAG system would actually hand to a generator: q3_two_hour_notification, semantic chunking, cosine 0.6814, pulled from fr_2024_ptasp_final_rule.txt when the answer exists only in ecfr_49cfr674_state_safety_oversight.txt.

The retrieved chunk is 2,955 characters of rulemaking commentary about Sec. 673.29(b) and whether a hazard requirement "applies to all relevant hazards or only hazards that meet a determined risk rating." It shares nearly the whole vocabulary of the query -- safety event, hazard, requirement, transit agency, FTA, notification -- while containing none of its answer.

That is the failure mode the embedding is built to have. MiniLM encodes topical similarity, not propositional content: "how quickly must an agency notify FTA of a safety event" and "which hazards require a risk rating" land in almost the same region of the space, because both are about transit-safety reporting obligations. The phrase that actually discriminates, *within two hours*, is one clause in a 38 KB regulation, and averaged into a 2,955-character chunk embedding it carries almost no weight. Nothing about the score is wrong; the chunk really is topically close. It just does not answer the question.

q3 is the corpus's hardest query by a wide margin: 10 of the 17 confidently-wrong retrievals belong to it, and it produced the lowest rank-1 cosine for all three techniques. The question was chosen because Part 673 (safety plans) and Part 674 (state oversight) discuss overlapping subject matter in near-identical register, and that overlap is exactly what the embedding cannot separate.

2.9 Observations

Sentence-window chunking won, and it won because its retrieval unit matches the size of the thing being retrieved. Every answer in questions.yaml is one or two sentences -- a probable-cause finding, a deadline, a percentage. The mean sentence-window node embeds 317.8 characters, against 1,980 for token and 2,146 for semantic. When a one-sentence answer is averaged into a 2,000-character chunk vector, the rest of the chunk dilutes it; at 318 characters it dominates. That shows up directly in the scores: top-1 cosine 0.7809 against 0.7532 and 0.7476, mean@5 0.7615 against 0.7193 and 0.7158, and the best rank-1 cosine on three of the five questions (q2, q4, q5). The window itself costs nothing at retrieval time -- the neighbouring sentences sit in metadata, excluded from the embedding, so each node carries a mean 1,151.9 characters of context, 3.6x its embedded length, without polluting the vector that does the matching.

Semantic chunking was the weakest and by far the most expensive, and the chunk statistics say why. Its chunks range from 3 characters to 51,993 -- against 197-3,506 for token. MiniLM truncates at 256 word-pieces, roughly 1,000 characters, so a 51,993-character chunk is about 98% invisible to the model that embeds it. The 95th-percentile breakpoint threshold, chosen to keep chunks topical, simply never fired inside the long enumerated passages that federal regulations are built from, and the result is a handful of enormous chunks whose embeddings represent only their opening paragraph. Semantic was the only technique to get a rank-1

answer wrong (q3), it owns 7 of the 17 confidently-wrong retrievals, and it cost 699.84s to chunk against token's 1.88s -- 372x the price for the worst retrieval quality of the three.

One caveat on Recall@5: all three techniques scored 1.00, so document-level recall does not discriminate between them at all on this corpus. Every technique put the right file somewhere in the top 5 for every question. The separation is entirely in rank-1 correctness and in cosine, which is why those columns carry the argument above. The best technique does vary by query -- semantic produced the single highest cosine in the whole run (0.8517 on q1, the Sacramento probable-cause question, where the answer is a long formal paragraph that its larger chunks happened to capture whole).

2.10 Conclusion

Sentence-window chunking is the best technique for this corpus. It has the highest top-1 cosine (0.7809) and mean@5 cosine (0.7615) of the three, it is the only technique with rank-1 correct on all five questions, and it wins the rank-1 cosine on three of them. The price is 2.5x the retrieval latency of the other two (235.14 ms against ~93 ms) because the store holds 6,008 vectors instead of ~500, and 510.16s to embed them -- real costs, but one-time at index build and still sub-second at query time. Semantic chunking is not worth its price here: it is the slowest to build by two orders of magnitude, produces chunks up to 51,993 characters that the embedding model can only partially read, and returns the worst retrieval quality of the three. Token chunking is the pragmatic middle -- within 0.03 cosine of the winner for a fortieth of the total build cost -- and would be the sensible default if index build time mattered.

Part 3 - Verification

code/verify_hw03.py runs the whole self-check and writes reports/hw03/verification.json. It confirms the six configuration values, starts the FastAPI app on port 8470 and drives the full auth flow over HTTP, re-hashes every corpus file against CORPUS_MANIFEST.json, validates questions.yaml, and checks the retrieval output and METRICS.md. It also compares the commit timestamp of questions.yaml against that of raw/retrieval_rows.json, so the "questions committed before results" requirement is evidenced by the repository rather than asserted in prose.

SCREENSHOT: 3.1 - make verify-hw03 summary

```
-- Open project in Codex | ritchit -- -zsh      ~/Desktop/DATA 260--Agentic AI/data260-3170 -- -zsh      ~/Desktop/DATA 260--Agentic AI/data260-3170/code/web_application -- -zsh      +

[PASS] bootstrap:alert_on_login
[PASS] bootstrap:cards

Part 1 - live auth flow on port 8470
[PASS] server:port_free -- 8470
[PASS] server:started -- uvicorn app:app on 8470
[PASS] route:home_200 -- GET / -> 200
[PASS] route:dashboard_anonymous_redirects -- 303 -> /login?expired=1
[PASS] route:login_200 -- GET /login -> 200
[PASS] auth:bad_password_rejected -- -> 401
[PASS] auth:bad_password_show_alert -- Bootstrap alert rendered on the login page
[PASS] auth:bad_password_issues_no_session
[PASS] auth:unknown_user_same_response -- does not reveal which half was wrong
[PASS] auth:good_login_redirects -- 303 -> /dashboard
[PASS] auth:session_cookie_issued
[PASS] cookie:httponly
[PASS] cookie:secure
[PASS] cookie:samesite
[PASS] cookie:all_three_attributes -- s3170_session=eyJ2aWQlIAWTlRYUZY24SHBnQk1X30tdEZNTjQkQkYhXkhWtWm1f0==.argISg.2oArV5o3isDHohucaI5wZ_1hkvQ; path=/; M
[PASS] cookie:header_captured_to_file -- reports/hw03/raw/set_cookie_header.txt
[PASS] route:dashboard_authenticated_200 -- GET /dashboard with session -> 200
[PASS] dashboard:greet_user
[PASS] route:logout_redirects -- 303 -> /
[PASS] session:logged_out_cookie_cannot_reach_dashboard -- replayed a valid signed cookie after logout -> 303
[PASS] session:relogin_ok
[PASS] session:live_before_timeout -- -> 200
[PASS] session:idle_timeout_blocks_dashboard -- after 3s idle -> 303 /login?expired=1
[PASS] hw2:api_still_works -- HW2 REST API unaffected by the HW3 changes
[PASS] hw2:form_moved_to_incidents

Part 2 - corpus and retrieval artefacts
[PASS] file_exists:fetch_corpus -- /Users/ritchit/Desktop/DATA 260--Agentic AI/data260-3170/code/fetch_corpus.py
[PASS] compiles:fetch_corpus
[PASS] file_exists:run_rag_chunking -- /Users/ritchit/Desktop/DATA 260--Agentic AI/data260-3170/code/run_rag_chunking.py
[PASS] compiles:run_rag_chunking
[PASS] file_exists:summarize_rag_metrics -- /Users/ritchit/Desktop/DATA 260--Agentic AI/data260-3170/code/summarize_rag_metrics.py
[PASS] compiles:summarize_rag_metrics
[PASS] file_exists:questions_yaml -- /Users/ritchit/Desktop/DATA 260--Agentic AI/data260-3170/reports/hw03/questions.yaml
[PASS] file_exists:source_md -- /Users/ritchit/Desktop/DATA 260--Agentic AI/data260-3170/reports/hw03/SOURCES.md
[PASS] file_exists:corpus_manifest -- /Users/ritchit/Desktop/DATA 260--Agentic AI/data260-3170/reports/hw03/CORPUS_MANIFEST.json
[PASS] corpus:present -- 8 files
[PASS] corpus:size_over_200kb -- 4,396,156 bytes (4293.1 KB), floor 284,880
[PASS] corpus:hashes_match_manifest -- 8 documents verified
[PASS] manifest:records_sha256
[PASS] questions:at_least_five -- 5 questions
[PASS] questions:have_expected_answers
[PASS] questions:have_expected_source_files
[PASS] questions:at_least_two_single_source -- 4 marked single_source
[PASS] questions:expected_files_exist_in_corpus -- all present
[PASS] git:questions_committed -- 2026-09-21T19:42:41+00:00
[PASS] git:questions_committed_before_results -- questions 1790019761 vs results 1790019761
[PASS] raw:retrieval_rows -- 75 rows
[PASS] raw:three_techniques -- semantic, sentence, token
[PASS] raw:csv_present
[PASS] raw:graded_embed_model -- embed_model=sentence-transformers/all-MiniLM-L6-v2
[PASS] metrics:written -- 174 lines
[PASS] metrics:no_placeholders
[PASS] metrics:confident_miss_found -- the assignment requires at least one confidently-wrong retrieval
[PASS] git:hw3_tag_present -- hw3

=====
75 passed, 0 failed, 0 skipped (of 75 checks)
wrote reports/hw03/verification.json
=====
e)
260-3170 on p main [!] via v3.12.13 (.venv) on [us-east-1] took 7s
>
```

The header of the generated file:

```
{
  "homework": "HW3",
  "sid4": 3170,
  "configuration": {
    "port_base": 8470,
    "prefix": "s3170",
    "domain_id": 2,
    "domain": "Municipal Transit Incidents",
    "idle_timeout_used_in_test_s": 3
  },
  "seed": 3170,
  "verify_seed": 263170,
  "total_checks": 74
}
```

AI_USE

Full version: `reports/hw03/AI_USE.md`.

1 What I used an AI assistant for, and what I did myself

Assistant: drafted `auth.py`, the four Jinja2 templates and the `SessionMiddleware` wiring; drafted `fetch_corpus.py`, `run_rag_chunking.py`, `summarize_rag_metrics.py` and `verify_hw03.py`; drafted the README HW3 section and this report structure.

Me: chose the eight-document corpus and the genre mix behind it; wrote and committed `questions.yaml` before any retrieval ran, checking every expected answer against the source document rather than a summary; downloaded the corpus and ran the pipeline on my own machine; took all screenshots and assembled this report; reviewed and corrected the drafted code - four of those corrections are below.

2 An AI-produced output that was wrong or unsuitable

- (a) The first auth implementation could not revoke a session (section 1.4): it treated a signed cookie as server-side state, and would have failed the requirement it was written for.
- (b) The recomputed cosine did not match the retriever's own score for sentence-window chunking, disagreeing by up to 0.18.
- (c) A Federal Register source URL was built by guessing the publication date from a search result and returned 404. Separately, the first dependency pins required a torch newer than 2.2.2, which does not exist for Intel macOS - the install would have failed on this hardware.
- (d) Submitting the login form with empty fields returned FastAPI's raw 422 JSON instead of the Bootstrap alert, because a required Form field treats an empty value as missing under Pydantic 2.12 and the framework's error handler fired before any of my validation code ran.

3 How I detected the problem

- (a) A smoke test written before trusting the code: log in, save the cookie, log out, replay the saved cookie in a fresh client. Expected a redirect, got 200 OK. That test is now permanent as `session:logged_out_cookie_cannot_reach_dashboard` in `verify_hw03.py`.
- (b) Printing the store score and the recomputed cosine side by side. They matched for token and semantic chunking and were consistently off for sentence-window only, which pointed at a difference between the techniques rather than at the arithmetic.
- (c) `fetch_corpus.py` refuses to save any response under 2,000 bytes, so the bad URL surfaced as a named failure rather than a silently truncated document. The torch pin was caught by resolving the full dependency set before installing it.
- (d) Found by hand while taking the screenshots for this report - clicking Log in on an empty form. It was then reproduced in a test client across four input cases before being fixed.

4 What I changed, and why it works now

(a) Sessions are revocable: a random session id per login, registered server-side, deleted on logout and on idle expiry. A replayed cookie fails at the registry lookup even though its signature is valid. Authority moved from the cookie to the server, which is what makes revocation possible at all.

(b) The re-embedding uses MetadataMode.EMBED, matching the string LlamaIndex actually indexed. The recomputed cosine now agrees with the store score to six decimal places for all three techniques.

(c) Federal Register documents are resolved through the public API by document number, which is stable, with the date-based URL kept only as a fallback; SOURCES.md records the URL actually retrieved. Dependencies are pinned to torch 2.2.2 / sentence-transformers 3.0.1 / transformers 4.44.2 / numpy<2, verified to resolve together on this hardware.

(d) The two Form fields default to an empty string instead of being required, so a blank submission reaches verify_credentials() and is rejected like any other bad credential, rendering the alert. novalidate was also removed from the form so the browser enforces its own required check first.

Appendix - files

Path	Contents
code/web_application/auth.py	Authentication router, user store, session registry
code/web_application/templates/	base, home, login, dashboard (Jinja2 + Bootstrap)
code/fetch_corpus.py	Corpus download, conversion, SOURCES.md and CORPUS_MANIFEST.json
code/run_rag_chunking.py	The three chunkers, indexes and the retrieval-only function
code/summarize_rag_metrics.py	Recomputes METRICS.md from reports/hw03/raw/
code/verify_hw03.py	Self-check; writes reports/hw03/verification.json
reports/hw03/questions.yaml	The five graded questions, committed before the run
reports/hw03/RUN_LOG.txt	Console output of the graded run
reports/hw03/raw/	retrieval_rows.jsonl/.csv, chunk_stats.json, query_vectors.json